

ICT302 IT Professional Practice Project

Project Name:

Secure QR Code Development

Client Organization:

Secure QR Code Project Client

Group Name:

VAFPQR

REQUIREMENT AND ANALYSIS

Submitted by:

Lim Yu Jun Felixia [redacted]

Submitted on: 1st Aug 2026

Version: 3.0

Name	Student ID	Signature	Date
Patricia Mikaela Marie Jusay Orenza	[redacted]		1st Aug 2026
Lim Yu Jun Felixia	[redacted]		1st Aug 2026
Roy Yoo Hojin	[redacted]		1st Aug 2026
Gu QiYu	[redacted]		1st Aug 2026
Alvin Rui Long Yeow	[redacted]		1st Aug 2026
Vanessa Tan	[redacted]		1st Aug 2026

Table of Contents

1. Executive Summary.....5
2. Introduction.....6
2.1 Purpose of the Project.....6
2.2 Background.....6
2.3 Problem Description.....6
2.4 Opportunities.....7
2.5 Project Objectives.....7
2.6 Development Methodology.....7
3. Proposed Solution Outline.....8
3.1 Solution Overview.....8
3.2 Project Scope.....9
3.3 Constraints.....9
3.4 Hardware and Software Environment.....9
4. Functional Requirements.....10
4.1 System actors.....10
4.2 Functional requirements matrix.....11
5. Non-functional Requirements.....18
5.1 Security Requirements.....18
5.2 Performance Requirements.....18
5.3 Reliability Requirements.....18
5.4 Availability Requirements.....18
5.5 Compatibility Requirements.....18
5.6 Usability Requirements.....19
5.7 Maintainability Requirements.....19
5.8 Assumptions.....19
6. Diagrammatic Representation of the Requirements.....20
7. Conclusion.....25
8. Appendices.....27
8.1 Appendix A – Deliverable Task Breakdown Statement (signed, scanned and inserted into document).....27
8.2 Appendix B - Copies of client documents on which you based the analysis.....28
8.3 Appendix C – Glossary.....28

Version	Date	Description of Changes
1.0	8 June 2026	Submitted first R&A
1.1	24 June 2026	Edited ES and formatting according to feedback
1.2	5 July 2026	Elaborated and finalised on functionality requirements after client meeting
2.0	21 July 2026	Finalised ES and updated ERD diagrams and descriptions
2.1	22 July 2026	Added version control, edited functionality and non-functionality requirements table after feedback from demo presentation
3.0	1 August 2026	Final Submission

1. Executive Summary

Conventional QR codes are susceptible to substitution, alteration, and redirection to malicious websites because they lack an integrated mechanism for authenticity verification. Because there is no simple way to identify or report hacked code, individuals, organizations, and event or advertising platforms are exposed to phishing and fraud.

The criteria, scope, and potential solution for a Secure QR Code Authentication System are outlined in this Requirement and Analysis document. Our clients are Peter Cole and Mike Groeneweg. The proposed system uses a JWT based authentication framework to address these security flaws. Before any redirection is allowed, each QR code contains a signed token with the expiration time, issue time, and destination information. This token is authenticated, including a blacklist check. The solution comprises a secure QR code generator, a QR validation API, a controlled redirect system, a mobile scanning application, a tampering warning reporting mechanism, an admin dashboard, and a supporting database management system.

The key functional features are:

- secure QR code creation
- JWT validation and expiry checking
- blacklist based revocation
- mobile scanning with tampering alerts
- dashboard based administrative monitoring

The key non-functional features are:

- performance (sub three second redirection, ≥99% generation/read success rates) dependability
- backward compatibility with current QR scanners
- ease of use requiring no specific training.

The functional and non-functional requirements, project scope and constraints, and suggested system architecture are described in this document, which serves as the analytical basis for the subsequent phases of design, development, testing, and deployment. If the solution is successfully implemented, the security risks associated with unauthenticated QR code use will be significantly reduced.

2. Introduction

2.1 Purpose of the Project

This project aims at creating an improved QR code system called the "Secure QR Code Development" in order to improve the security of QR codes for public use. In this project, the QR code would be authenticated before any redirection to their destination while providing ways of detecting and reporting any altered QR codes.

2.2 Background

Currently, there are several uses for QR codes including advertisement, payment processes, event registrations, and so on. Despite the fact that QR codes are easy to use, they are open to manipulation and replacements in public spaces.

It could easily be manipulated by a malevolent individual such that when scanned, a user is redirected to a phishing website or any malicious content. Currently, QR codes do not have a method that can authenticate QR codes before any redirection occurs.

This is where the "Secure QR Code Development" project comes in to resolve this problem.

2.3 Problem Description

Conventional QR codes offer no means for verification of the genuineness of the QR code being used. Users usually tend to scan QR codes without being aware of whether they have been altered in any way.

The major problems that arise include:

- QR code can easily be substituted
- Users can be directed to harmful sites using QR codes
- Limited monitoring of QR code usage by organizations
- No quick means for suspicious QR codes to be reported
- Conventional QR codes offer inadequate measures for prevention of frauds

2.4 Opportunities

This project presents the following opportunities:

- Improving security and trust in the use of QR codes
- Minimizing risks related to QR code substitution
- Providing organizations with analytical data on the use of QR codes
- Making suspicious QR codes reportable quickly
- Ensuring confidence in the interaction with QR codes by people

2.5 Project Objectives

The objectives of the Secure QR Code Development project include:

- Developing a QR code generation framework that can ensure security
- Creating a framework that can authenticate and validate QR codes
- Implementing an expiry control functionality for generated QR codes
- Developing a framework for QR code management
- Creating a mobile application to validate QR codes
- Implementing a telemetry and metrics functionality
- Creating a tamper detection and reporting functionality
- Ensuring that the framework is compatible with standard QR code readers

2.6 Development Methodology

This project will adopt the iterative development methodology. The requirements for the development of the project will be obtained from the client, analyzed, and then implemented in several development and testing iterations.

3. Proposed Solution Outline

3.1 Solution Overview

The proposed solution is called Secure QR Code Authentication System which integrates various solutions for protecting the users from malicious QR codes.

Unique QR codes can be generated with the help of identifiers and secure authentication methods. In case a QR code is scanned by a user, it will undergo the process of authentication in order for access to the respective web page to be possible.

The system has the following major components:

1. **Secure QR Code Generator**

   The QR Code Generator generates unique QR codes that include JWT-based authentication tokens and UUIDs. Security information such as expiry times, issue times, authentication credentials, etc. can also be included in the QR code.

2. **QR Validation API**

   The QR Validation API performs a check of QR codes in terms of signature validation, expiration times and whether the QR code is blacklisted or not. If a QR code is invalid or expired, access will not be granted.

3. **Redirect System**

   With the help of this component, access to the destination URL will be controlled. The user will be redirected only if the QR code is validated properly.

4. **Mobile Application**

   This application is used for scanning, verification, and reporting of suspicious QR codes.

5. **Tampering Alert Reporting**

   The reporting system allows users to create reports where suspicious QR codes are reported. Reports can consist of GPS locations, images, description, and any other contact information if required.

6. **Admin Dashboard**

   The admin dashboard system allows the administrators to manage and manipulate QR codes, view reports and alerts, access metrics, and blacklist the compromised QR codes.

7. **Database Management System**

   The database will be used to store all the required data about QR codes, log-in, telemetries, reports, and alerting.

3.2 Project Scope

This project entails:

1. Creation and management of QR codes
2. Authentication of QR codes
3. Expiry date management of QR codes
4. QR codes scanning on mobile devices
5. Telemetry data collection from the usage of QR codes
6. Alert reports about tampering detection
7. Dashboard management for administration purposes
8. Blacklist compromised QR codes

This project does NOT entail:

- Payment processing capabilities
- Using external third-party authentication services
- Physical installation and maintenance of QR codes
- Any large-scale hosting service for QR code

3.3 Constraints

There are various constraints in place for the project:

1. Short timeline of the project
2. Limited development resources
3. Constraints in academic projects
4. Compatibility issues with existing QR scanners
5. Limitations of mobile devices' compatibilities
6. Security requirements of the client

3.4 Hardware and Software Environment

The selected approach would involve the use of:

Hardware:

- Android smartphones
- Workstations for development
- Server architecture hosted on the cloud

Software:

- Node.js back-end services
- Database system management
- GitHub platform
- Microsoft Teams and OneDrive platforms
- Android app developer tools
- JWT authentication modules
- QR code generating modules

4. Functional Requirements

4.1 System actors

Administrator:

A verified system administrator responsible for generating secure QR codes, modifying token operational statuses through the management interface and through a specialized control panel anlyze the system performance.

User:

Unauthenticated individuals who scan the QR codes using the built-in device camera features or a specific mobile app initiating automated system redirects and sends structured tampering information.

4.2 Functional requirements matrix

ID	Functional requirement	Description	Criticality	Technical issues	Cost and schedule	Risks and mitigations	Dependencies	Acceptance criteria
FR-01	Secure QR code generation	System will produce distinctive cryptographic QR codes containing a JSON Web Token (JWT) including structured expiration settings, foundational destination routing URL and timestamp. JWT token will be encoded as data payload within the QR code.	High	Execution of cryptographically secure signing methods, securely isolating JWT signing keys from the standard application processes and making sure token collision resistance during simultaneous generation requests.	12 hours	Risk: Vulnerability or breach of private server-side encryption configuration keys. Mitigation: Implement strict server-side storage of validation keys in external environment configuration files (.env) that are not included in version control system.	Database system initialisation	System shall generate a QR code image that encodes a valid JWT token payload which enables successful verification and decoding.

FR-02	QR code validation and authentication	System analyses incoming JWT payloads to assess the structural validity of the cryptographic signature, aid in preventing replay-based misuse through token validation tests, verify expiry time and check the database blacklist table before granting redirect access.	High	Handling query delays and verification logic amid simultaneous database accesses during peak request periods along with overall performance enhancement factors.	15 hours	Risks: Delays in network connectors slowing down signature verification processes during mobile operations with low bandwidth. Mitigation: Perform JWT signature validation and payload integrity assessments in memory before the initiation of regular database queries.	FR-01, database system	System shall refuse QR tokens that have invalid or tampered signatures.
FR-03	Redirect system	System securely directs the validated user to their designated destinations while intercepting and redirecting invalid traffic to warning screens if any tokens are identified as expired or blacklisted.	High	Applying URL sanitization and server-side verification for the permitted destination domains.	5 hours	Risk: Manipulation of redirection paths caused by parameter injection or the creation of malicious strings. Mitigation: Maintain a server-side setup of authorized destination targets.	FR-02	System shall redirect standard user requests to the valid intended destinations and invalid or blacklisted traffic to a warning interface.

FR-04	Mobile application	Application will connect with the built-in camera hardware to obtain camera input for QR code scanning then analyses the deployment links and send verification requests along with the device metadata and to the backend API validation layer.	High	Managing camera permission lifecycles across different platforms, addressing connection timeouts smoothly and enhancing frame processing efficiency.	20 hours	Risk: Occasional software instability caused by fragmentation of system devices across different operating system. Mitigation: Uphold a unified cross-platform logic standard that is verified across various emulated platforms by employing React Native code abstractions.	FR-02, backend API	System shall automatically conduct a successful scan-to-request in order to obtain the target string and creates valid outbound verification requests.
FR-05	Mobile Session Security	The mobile client (supabaseMobile) must use Expo SecureStore to securely store JWTs throughout the app sessions.	High	Overseeing secure platform-native secure enclave storage on various operating systems, effectively address token retrieval delays and maintaining encrypted session	15 hours	Risk: Man-in-the-middle (MitM) attacks can happen easily as the data traffic would be vulnerable. It also allows for 3rd parties to commit identity spoofing. Mitigation: Enforce HyperText Transfer Protocol Secure (HTTPS) and the use of Json Web Token (JWT) during the transactions of data	Backend API, database system	The use of HTTPS and JWT is detected when reading the data traffic between a user and the system.

				continuity.		between the user and the system.		

FR-06	Blacklist control	Stops additional routing attempts by enabling instant token containment features and permitting operators to change token's status.	High	Handling the propagation delay of state changes across ongoing application sessions and addressing real-time data consistency issues.	6 hours	Risk: Minor delay in immediate propagation of the blacklist caused by delays in database indexing or caching conditions. Mitigation: Implement indexed database queries and direct cache invalidation methods to guarantee immediate updates to the blacklist.	FR-02	Blocking any further routing attempts for future scans by setting a particular item's status to 'blacklisted'.
FR-07	Admin dashboard	Web administration module featuring a secure visual control panel that is safeguarded by role-based access control enabling only authorized users to set up new tokens, change system blacklist and access visual metrics.	Medium	Ensuring consistent state across backend REST endpoints and reducing the risk of session hijacking for admin profiles via session mechanisms that are secure.	18 hours	Risk: Delays in key project milestones caused by the complexity of front-end dashboard structure surpassing the allotted time. Mitigation: Reduce interface complexity by employing modular front-end elements and gradual development.	Authentication system, FR-02	Authorized administrative accounts will generate, modify and oversee all blacklist entries and QR code records using the online interface.

FR-08	Tampering report system	End-users will be allowed to gather and send suspicious QR code findings which includes the description, location and image media proof through a server-side file verification procedure.	Medium	Guaranteeing the comprehensive effectiveness of database functions during the processing and validation of multi-part media storage.	15 hours	Risk: Data reporting processes interruption due to denial of hardware permissions (GPS/camera). Mitigation: Setting up a logical user interface backup option which allows users to manually enter location details if collection from device sensors fails.	Backend API, database system	System shall accept multipart report submission; store verified report information in the database and enforce a limit for file size of 5MB.
FR-09	Scan analytics	Telemetry system documents the incoming log scan metadata that includes timestamp, locations and device characteristics for system monitoring and management.	Low	Reducing database I/O transactional strain during periods of intense simultaneous traffic while adhering to privacy protection principles and data minimization.	7 hours	Risk: Large-scale logging ingestion that consumes persistent server storage faster than expected. Mitigation: Introduce sequential date-monitoring variables and employ space-efficient column data types.	FR-02, database logging system	System shall log incoming data attributes to analytical tables during standard load conditions without impacting response time.

Legend

High = Critical, the system cannot function securely or without it.

Medium = non-critical, it is core magement or operational feature but without it, the system can still conduct baseline functions.

Low = Secondary / optional

5. Non-functional Requirements

5.1 Security Requirements

- **Secure communication:** System will utilize secure, platform-controlled HTTPS endpoints for all client-server communications.
- **Authentication:** The system will use platform-verified JSON Web Tokens (JWT) to manage sessions, incorporating automatic token rotation.
- **Secure storage:** Confidential session data needs to be safeguarded with secure storage appropriate to the platform (e.g., Expo SecureStorage for mobile and localStorage for web) to prevent any unauthorized access.
- **Access control:** Role-based access control (RBAC) enforced at the database level with Row-Level Security (RLS) policies, ensuring that data access is restricted based on user roles and defined security protocols.
- **File integrity:** Every file uploaded must be validated and sanitized to prevent the injection of any malicious content.

5.2 Performance Requirements

- **Redirect speed:** Redirection by the system should take place after a maximum of three to five seconds after the validation of the scanned QR code.
- **QR generation accuracy:** Success rate of QR Code generation must be not less than 99%.
- **Reading accuracy:** QR Code reading success rate must be not less than 99%.
- **Validation latency:** Real-time QR code validation request is an expected functionality.

5.3 Reliability Requirements

- **Backup strategy:** The system should carry out daily database backups to ensure that data will not be lost, allowing for reliable point-in-time recovery.
- **Failure recovery:** Recovery measures should be carried out after the occurrence of failures within the system.
- **Data integrity:** Guaranteed to have zero data corruption while the system is operating normally, making sure that every database transaction is either completed or entirely rolled back to preserve a consistent data state.

5.4 Availability Requirements

- **Availability:** The backend infrastructure during normal operating hours should always be available.
- **Monitoring:** Monitoring tools will be used to detect any disruption or anomaly in the service by providing live view into the backend infrastructure so to ensure constant accessibility.

5.5 Compatibility Requirements

- **QR standard:** There should be backward compatibility with QR code scanners. When a standard Android camera scans it, they will be redirected to an APK download link while users who are using the app will be redirected to intended URL.
- **Support across multiple platform:** There will be compatibility with multiple versions of Android devices, making sure that core scanning and SDK features work consistently across the older and current Android API versions.

- **Network resilience:** The system should work well across various networks managing fluctuating latency cellular data (3G, 4G, 5G) and Wi-Fi connections smoothly without losing application states.

5.6 Usability Requirements

- **Interface design:** The user interface must follow intuitive design principles to ensure that it is easy for all users to use.
- **Scanning workflow:** The scanning procedure will be simple and requires no specialized user training including a distinct on-screen framing guide that activates immediate automatic capture whenever the viewport detects a QR code.
- **Reporting efficiency:** The system must not be too demanding on user involvement to fill and send reports.

5.7 Maintainability Requirements

- **Extensibility:** Future developments should be possible within the system architecture, making certain that new database tables, mobile functionalities or APIs can be added effortlessly without causing any disruptions to the existing core systems.
- **Version control:** Thorough documentation and strict version control should be preserved for all source code, utilizing organized Git repositories and detailed API documentation to ease the onboarding process for the upcoming developers.
- **Modular architecture:** System components should be modular for easy modifications in the future, dividing backend routes, mobile UI views and authentication hooks into distinct, independent layers.

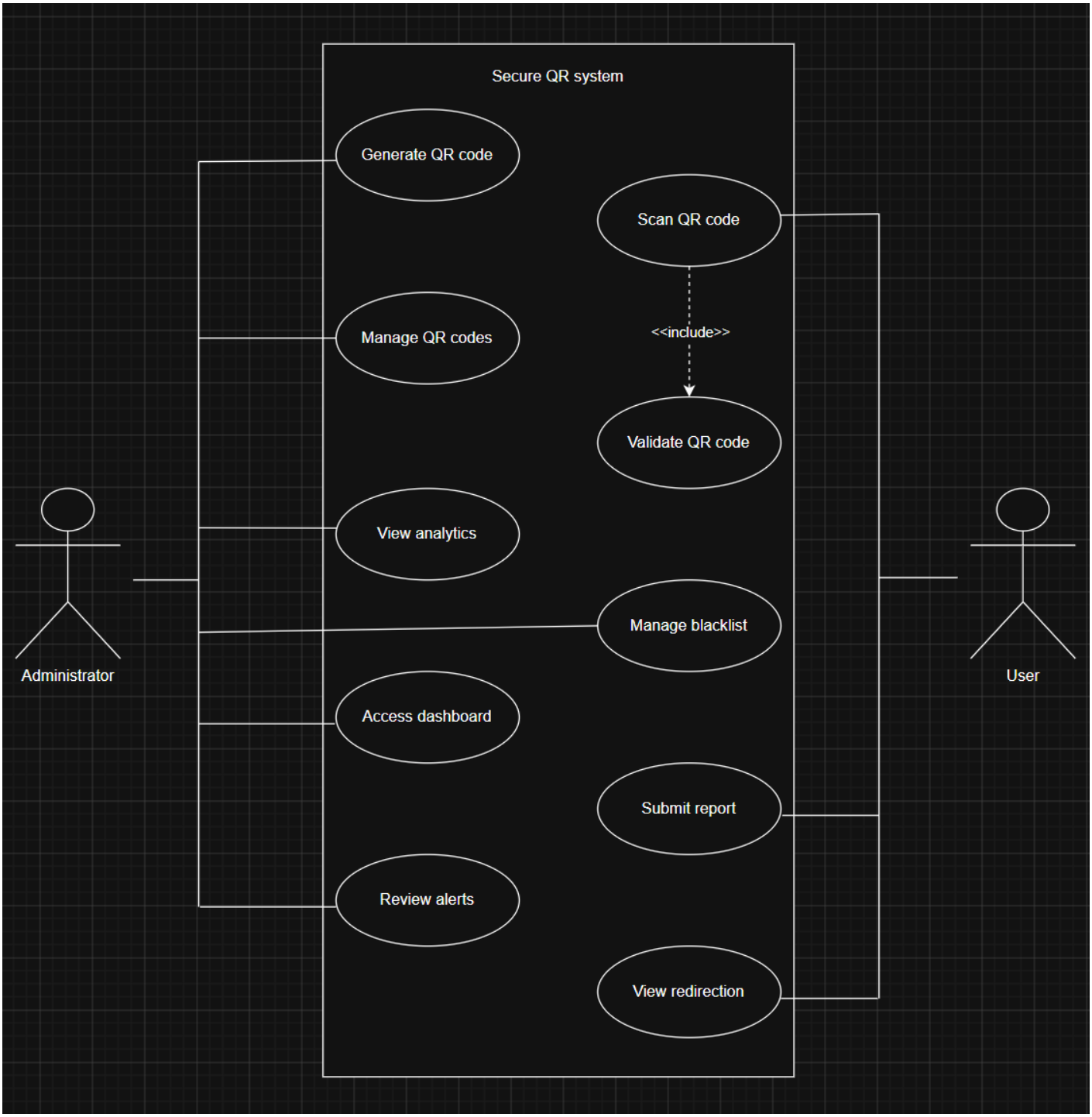
5.8 Assumptions

The following assumptions have been made:

- **Device compatibility:** It is assumed that users own Android devices that can scan QR codes.
- **Network connectivity:** It is assumed that users have internet connection (cellular data or Wi-Fi) when they need to verify the QR Code with the backend server.
- **Permissions:** Assuming users provide necessary permissions like camera access for scanning and location access for tracking.
- **Administrative competency:** It is assumed that the administrator**s** have the basic technical knowledge that is required to change roles, monitor alerts and manage data on the dashboard.
- **Infrastructure continuity:** Assuming that cloud hosting services, database provisions and backend server will be available throughout the project duration.

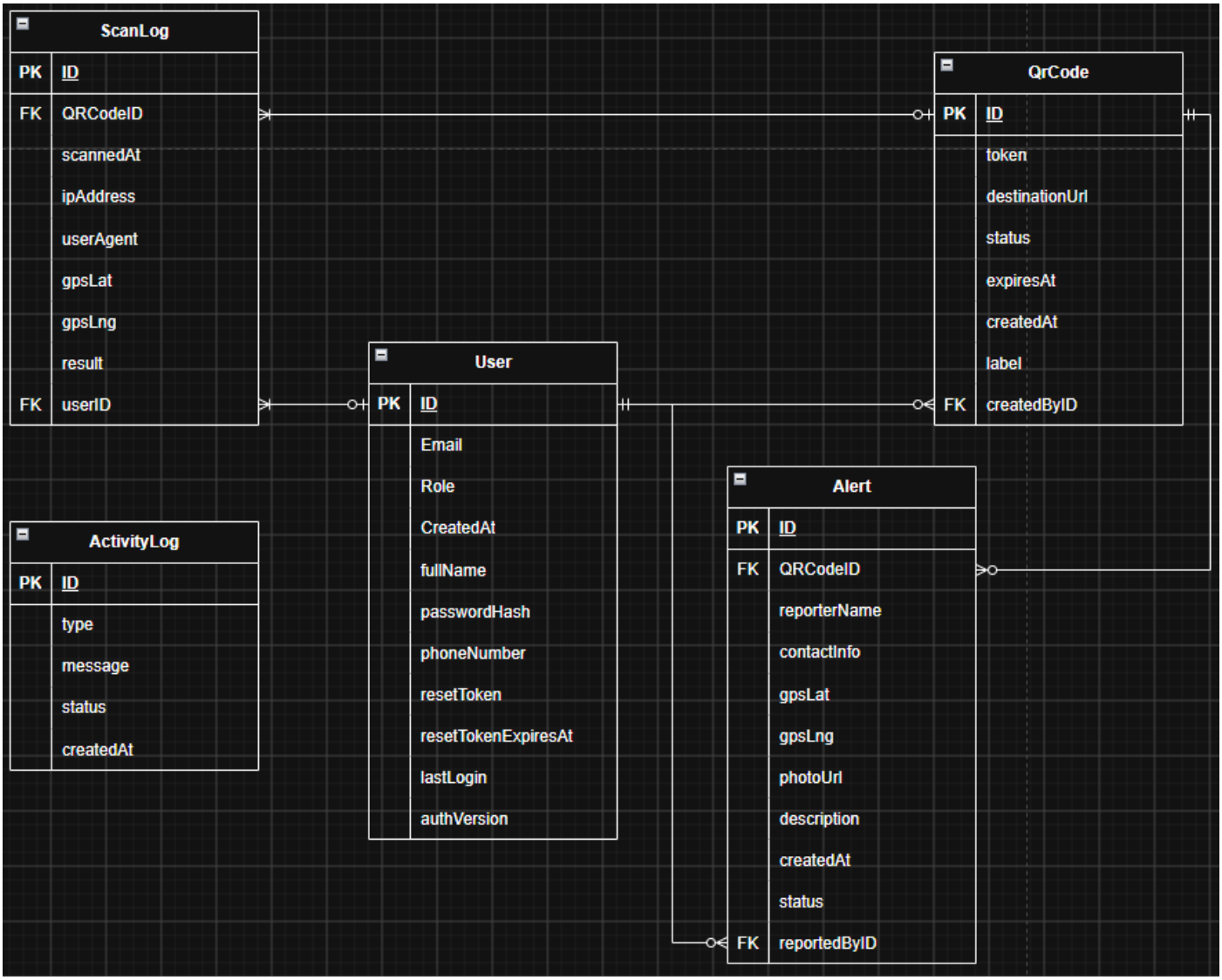
6. Diagrammatic Representation of the Requirements

Use Case Diagram



Based on the diagram, the main actors that will be making use of our Secure QR system are the administrators and the mobile app users. The administrators will be using our system to create and manage the QR codes made, manage the reports of the QR codes being tampered with and view analytics of the usage of the QR codes. Meanwhile, the users will be using our system to have reliable access to secure QR codes generated by the administrators and submit reports of any QR codes that have been tampered with. There will also be indications from the system during the scanning of the QR code to inform the user that the QR code is safe.

ERD Diagram



This ERD diagram shows the use of five tables: User, QrCode, ScanLog, Alert and ActivityLog.

The User table represents the Users and Administrators of the system. They are differentiated based on their roles given in the ‘Role’ attribute. The ‘resetToken’, ‘resetTokenExpiresAt’ and ‘last Login’ attributes in the table are related to the encryption part of our system. The Primary Key in this table is ID.

The QrCode table represents the QR codes that are tracked within the system. The ‘status’ attribute in the table will assign certain statuses tied to the QR code such as whether they are ‘active’ or ‘suspicious’ etc. The ‘createdAt’ and ‘expiresAt’ attributes show when the QR code was created and when it will expire. The Primary Key in this table is ID.

The ScanLog table represents the individual logs of the scans done through the system. The system will be capable of end-to-end scan tracking from scans done by invalid users to scans done on invalid QR codes by valid users. It will also be able to track the GPS locations of the scans done in the system. The Primary Key in this table is ID.

The Alert table represents the alerts of the QR codes being tampered with sent by users of the system. All the attributes are functions of the table. Examples include ‘gpsLat’ and ‘gpsLng’ showing the GPS location of the scan and ‘photoUrl’ displaying the URL of the photo etc. The Primary Key in this table is ID.

The ActivityLog table represents the recent logs of the activities which will be shown on the admin dashboard. The Primary Key in this table is ID.

Relationships:
User (1) → (0..*) QrCode
An administrator can have zero to multiple QR codes, while a QR code can only be generated by one administrator. This relationship allows the system to identify which administrator generated a specific QR code while also supporting accountability and auditing of QR code management activities.

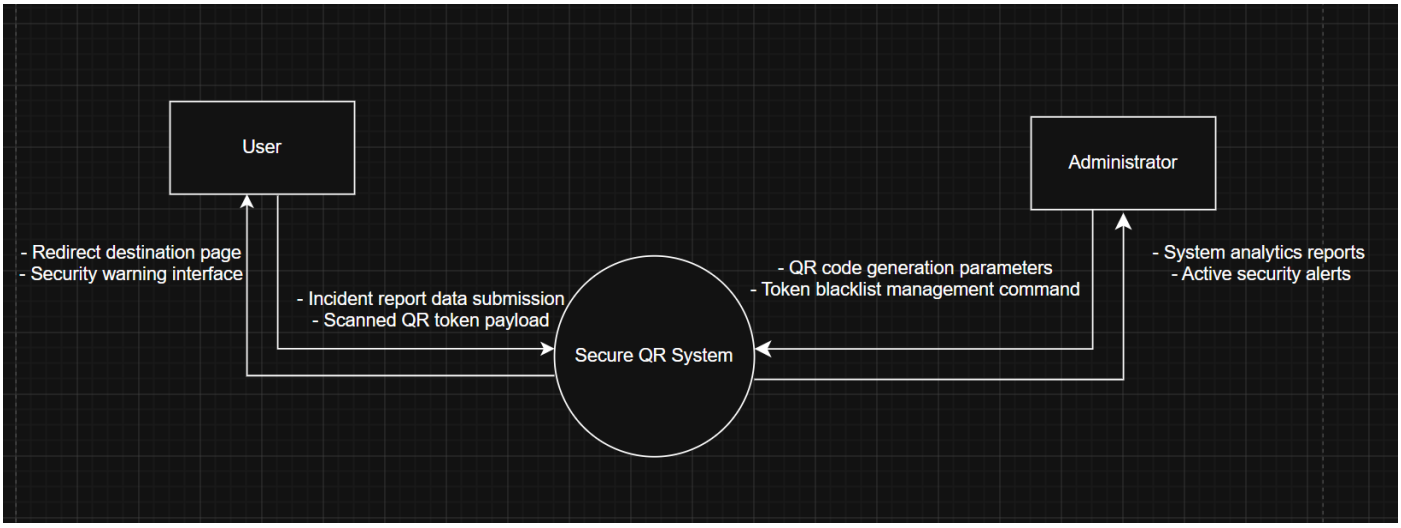
QrCode (0..1) → (0..*) ScanLog
A QR Code can have zero to multiple scan logs, while a scan log will always refer to a QR code. The only exception is if the QR code is deleted. This supports telemetry collection including the scan time, device information, validation results, location and IP address which can be used for analytics monitoring or security investigations.

QrCode (1) → (0..*) Alert
A QR Code can have zero to multiple tamper alerts, while a tamper alert will always refer to a QR code. This allows the administrator to examine all reports linked to a specific QR code and determine if the QR code should be blacklisted.

User (1) → (0..*) Alert
A user can report zero to multiple tamper alerts, while a tamper alert will always refer to a user. This allows the administrator to trace the reports back to the user who reported it.

User (0..1) → (0..*) ScanLog
A use can scan zero to multiple QR codes while a scan will always refer to a user. The only exception is if the user is deleted. This allows the administrator to trace the user who scanned the QR code.

Context Diagram

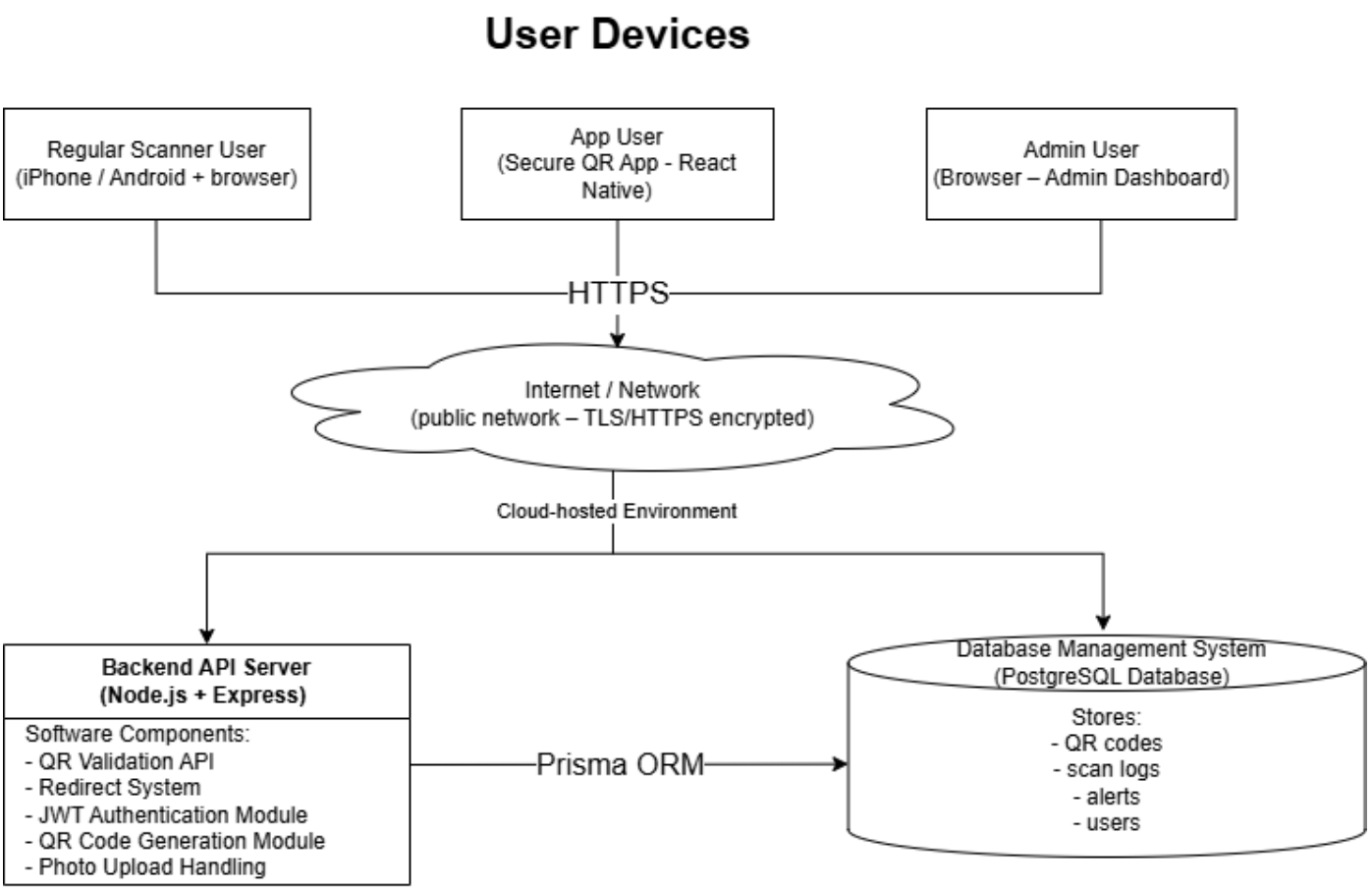


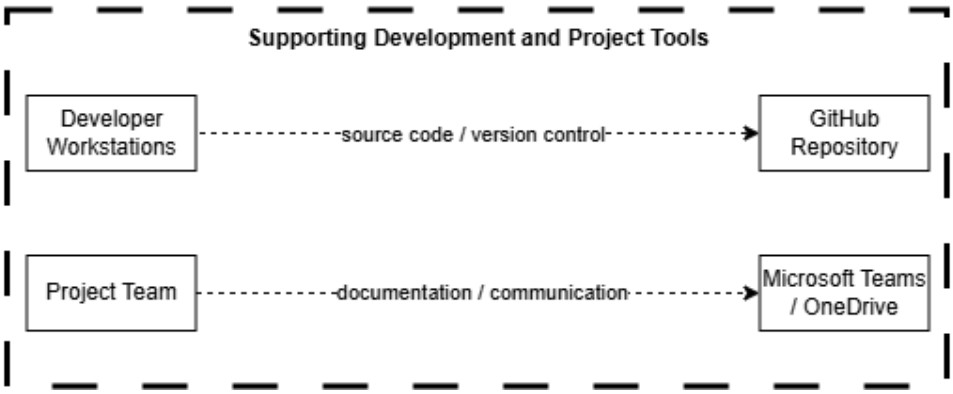
The Context Diagram shows the interactions between the external actors who are the User and Administrator and the system which is the Secure QR System.

The User interacts with the system by scanning the QR code which sends the QR token payload to the system to go through validation checks. Depending on the QR code, the system will then either send a redirect prompt to the user or send a warning message. The user can also send incident reports to the system about the QR codes being tampered with.

The Administrator interacts with the system by generating QR codes and performing QR Code and User management functions with the system. The system can also show metrics about QR usage to the Administrator and send tamper alerts to the Administrator.

Main Operational System





7. Conclusion

This Requirements and Analysis document has presented the business problem, project goals, system requirements, and the suggested system for solving the Secure QR Code Development project. The suggested system offers secure QR code creation, verification, monitoring, and detecting tampering functionalities to overcome the security weaknesses associated with conventional QR codes.

The collected requirements will serve as a solid starting point for the design, development, testing, and deployment process of the system. The successful completion of the project will result in improved security of QR codes.

Requirement	Verdict	Evidence / Notes
FR-01 Secure QR generation (JWT: expiry, URL, timestamp; keys in .env)	Met, exceeded	tokenService.js; QR_JWT_SECRET and AUTH_JWT_SECRET are separate secrets - stronger than the document requires.
FR-02 Validation & authentication (signature, expiry, blacklist checked before DB)	Met, exceeded	verifyService.js: JWT verification runs before any database query; authVersion adds replay revocation.
FR-03 Redirect system	Changed / partial	Browser users never reach the destination (B1). The documented destination allowlist is not implemented (D7); risk low because generation is admin-only.
FR-04 Mobile application	Met (except release)	Scan, verify, five result statuses implemented; only the EAS APK release step remains.
FR-05 Blacklist control	Met	A status change takes effect on the very next scan (verifyService.js).
FR-06 Admin dashboard (RBAC)	Met, exceeded	requireAdmin guards every admin route; last-admin guard and self-role-change block are extra protections.
FR-07 Tamper reporting (multipart, 5 MB limit)	Met (fallback gap)	5 MB cap and image-only MIME filter implemented exactly; GPS manual-entry fallback not implemented (D7).
FR-08 Scan analytics	Met, exceeded	Every scan including failures is logged; four-range bucketed charts exceed the documented scope.
NFR: redirect within 3 seconds	Conditional	Logic meets the target; Render cold start can violate it on the first request (D5). Pre-warm /api/health before demos.
NFR: backups, recovery, monitoring	Delegated / partial	Backups delegated to managed Supabase; the 3-minute downtime alert is not implemented (D8).

NFR: HTTPS, bcrypt, RBAC, upload validation	Met	All verified in code review.

8. Appendices

8.1 Appendix A – Deliverable Task Breakdown Statement (signed, scanned and inserted into document)

Deliverable Name:

Project Team Name	Team Number
VAFPQR	FT02

Name	Student ID	Deliverable Percentage Completed	Signature	Date
Patricia Mikaela Marie Jusay Orenza	[redacted]	10%		1 Aug 2026
Lim Yu Jun Felixia	[redacted]	18%		1 Aug 2026
Roy Yoo Hojin	[redacted]	18%		1 Aug 2026
Gu QiYu	[redacted]	18%		1 Aug 2026
Alvin Rui Long Yeow	[redacted]	18%		1 Aug 2026
Vanessa Tan	[redacted]	18%		1 Aug 2026

8.2 Appendix B - Copies of client documents on which you based the analysis.

Word Document:

ICT302 Secure QR Code Development TM 2026

Client:

1. Peter Cole
2. Mike Groeneweg

8.3 Appendix C – Glossary

Term	Definition
API (Application Programming Interface)	A set of policies and guidelines enabling the interaction and exchange of information among various software programs.
Authentication	The procedure of checking the validity and authentication of a QR code, user, or system component prior to granting access to the resource.
Backend	The back-end application of the system processing user requests, managing data, and executing validation procedures.
Blacklist	An enumeration of the QR codes that have become invalid, compromised, expired, or unsafe.
Dashboard	The web dashboard for tracking events in the system, handling QR codes, alerts, and monitoring analytics.
Database	A collection of data used for the storage of QR code information, user information, scan logs, and tampering incidents.
Expiry Date	The date after which a QR code will become invalid, unable to pass through the authentication procedures.
GPS (Global Positioning System)	A GPS-like location service used to establish the geographical coordinates of the user's mobile phone.
HTTPS (Hypertext Transfer Protocol Secure)	The protocol for securing data transmission between users and the server.
JWT (JSON Web Token)	The method of securely transmitting tokens carrying user information between system components.
Mobile Application	The Android program created for scanning QR codes and logging tampering incidents.
QR Code (Quick Response Code)	A two-dimensional barcode that carries

	information encoded into it for scanning by users.
QR Code Authentication	Process by which authenticity of a QR code is checked and ensured that it hasn't expired or gone through modification or blacklist.
Redirect System	An element used to direct the user to a destination website URL upon successful QR code verification.
Scan Log	A database storing data on scanning QR codes, consisting of date, time, location, and device information.
Telemetry	System information on the usage of QR codes and scanning activities conducted.
Tampering Report	A feedback report provided by the users in case QR codes seem suspicious, damaged, modified, or fake.
URL (Uniform Resource Locator)	The web address where users would be directed by a QR code after verifying successfully.
UUID (Universally Unique Identifier)	A special number created by the system to make every single QR code unique.
Validation	Checking the legitimacy of the scanned QR codes based on the required criteria, including expiration and being blacklisted.
Version Control	A tool used to track and manage modifications in the source code and project documents.
GitHub	Cloud-based services used by the development team for management and versioning of source codes and project documents.
OneDrive	Microsoft's cloud services utilized by the development team for collaboration in terms of source codes and documents.
User Acceptance Testing (UAT)	A type of testing carried out by stakeholders or users to see if the solution fulfills their business needs.
Secure QR Code System	The final solution implemented in this project for handling QR codes.